Mingke Wu

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EDUCATION

Peking University, Beijing, China 09/2021 – 07/2026

Bachelor of Medical Sciences (five-year program)

- Overall GPA: **3.71/4.0** (**88.32/100**), Rank **9th** out of **98**
- Final Defense for Bachelor's Degree: Rank **3rd** out of **98**
- Relevant Computational Coursework:
  Peking University: Introduction to Computation: **90**; Biomedical Data Processing and Biostatistics (MTTM I): **94**; Scientific Programming in R and Python (MTTM II): **97**; Applied Computational Methods in Biomedicine (MTTM III): **90**
  Johns Hopkins University Online: Linear Algebra (Linear Systems and Matrix Equations)
  Stanford University Online: Introduction to Statistics, Algorithms Specialization (4 courses: Algorithm design paradigms, Randomized algorithms, Graph algorithms and data structures, NP-completeness)

AWARDS AND SCHOLARSHIPS

Excellent Graduate Award (**top 2**)	07/2026
Grant for International Short-Term Exchange (**top 3**)	10/2025
Award for Scientific Research (**top 5**)	10/2025
Merit Student Award of Peking University (**top 5**)	12/2024
Outstanding Medical Student Scholarship, Second Prize (**top 5**)	12/2024
Merit Student Award of Peking University (**top 5**)	12/2023
Outstanding Medical Student Scholarship, Third Prize (**top 10**)	12/2023

PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Wu M**, Lu J, Ge Q. Sparse integrative dictionary learning resolves conserved drug-response states in multiple myeloma (MM). ***Cancer Genetics***. 2026;306–307:240–255. doi: 10.1016/j.cancergen.2026.08.002.
2. **Wu M**, Zhou Y, Xu D. Systematic evaluation of spatial transcriptomic annotation methods reveals conserved tumor microenvironment programs in NSCLC. ***Frontiers in Bioinformatics***. 2026;6:1880333. doi: 10.3389/fbinf.2026.1880333.
3. **Wu M***, Xu D*, Dou X, Zhou Y, Lu J, Ge Q. From structure to function: 3D tumor models for reconstructing tumor microenvironment regulatory networks and mechanical properties. ***Cell Signaling, Inflammation & Disease***. 2026;1:4. doi: 10.1186/s44505-026-00004-y.
4. **Wu M**. SPADE: An R package for spatial proximity analysis of differential expression. ***Bioinformation***. 2026;22(7):3954–3962. doi: 10.6026/973206300223954.
5. Wang J*, **Wu M***, Xu D, Jin R, Hao J, Wang X, Ge Q. An integrated single-cell transcriptomic atlas of human tricuspid and aortic valve diseases. ***Journal of Biosciences and Medicines***. 2026;14:360–392. doi: 10.4236/jbm.2026.145025.

Preprints & Under Review

1. Guo J, Liu Y, Ge X, Dou X, He Q, Zhou J, **Wu M**, Shao D, Wang Y, Zhao W, Wang Z, Liu C, Hao J, Lu D, Jin R, Chang Y, Lu J, Ge Q. A non-canonical MDK-SDC1-AKT axis controls SMAD2/3 activation to maintain proteostasis and apoptotic thresholds in plasma cell disorders. ***Chinese Medical Journal***. In press.
2. **Wu M**, Wu H, Zhou Y, Cheng S. SCAPE: An AI-Driven Platform for Comprehensive Single-Cell Data Analysis. ***bioRxiv***. 2025. Preprint. doi: 10.1101/2025.11.15.688526.
3. Ge X, Guo J, Liu Y, Dou X, He Q, Xu D, **Wu M**, Hao J, Wei Z, Xi W, Rong J, Lu J, Ge Q. Prior autoimmune disease reshapes the clonal plasma-cell transcriptome and remodels the bone marrow microenvironment in immunoglobulin light-chain amyloidosis. ***Advanced Science***. Revision submitted.

RESEARCH EXPERIENCE

Department of Genetics, Yale University 07/2026 – 10/2026

Remote Research Intern, full-time

Advisor: Dr. Monkol Lek

Pediatric Skeletal Muscle FAPs Atlas Construction via Deep Generative Latent Representation Learning

- Collected 60 pediatric skeletal muscle single-cell multiome datasets spanning infancy to adolescence, leveraging deep generative latent representation learning (MultiVI and MrVI) to construct a developmental muscle atlas.
- Identified age-associated compositional remodeling characterized by increased $CD55^{+}$ FAPs and decreased $GPC3^{+}$ FAPs.
- Revealed age-dependent functional remodeling during pediatric FAP maturation, with increasing lipid metabolism and TGF-β signaling consistent with adult FAP aging, but opposite age-associated trends in Wnt and VEGF pathways.
- Curated known FAP-derived signals that promote skeletal muscle development through MuSC receptors, then assessed their age-associated changes in pediatric cells to reveal potentially maturing FAP-MuSC signaling programs.
- Developed a cross-species representation alignment and transfer learning pipeline to validate age-associated shifts in FAP composition across external human and mouse datasets, and further investigated functional shifts in the learned latent space.

Department of Genetics and Urology, Yale University 04/2025 – 08/2025

Visiting Student, full-time

Advisors: Dr. Siyuan (Steven) Wang and Dr. Ping Mu

AI Agent-based Spatial Transcriptomics Analysis

- Leveraged Spatial Transcriptomics Agent to analyze RAEFISH datasets from adult mouse liver, lymph node and placenta, including differentially expressed genes, cell identity scores, and key genes within cell interactions.
- Quantified spatial zonation patterns, where RAEFISH revealed a clear zonation pattern in the liver and lymph node, and a pronounced maternal-fetal axis in the placenta, confirming RAEFISH's ability to capture spatial information.
- Integrated literature-based modules to interpret biological significance, identifying upregulation of *MHC* class II genes and *Cd74* in cholangiocytes near leukocytes, supporting their role as non-professional antigen-presenting cells.
- Found these results consistent with manual analyses and literature searches, demonstrating the reliability of the AI agents.

Genetic Perturbation Prediction Framework in Spatial Transcriptomics

- Trained genetic perturbation prediction models on public mouse data using linear (*State*) and deep learning (*CPA, scLAMBDA*) approaches, and applied them to spatial transcriptomics data, revealing lower concordance and highlighting the potential for future models that preserve spatial information.

LLM-Guided Cross-Model Transcriptomic Discovery of Lineage Plasticity in Prostate Cancer

- Dissected prostate cancer lineage plasticity across human, mouse, and cell-line models using LLM-derived scRNA-seq embeddings, uncovering and validating *E2F1* and *HMGB2* as mediators of enzalutamide resistance.

SCAPE: An AI-Driven Platform for Comprehensive Single-Cell Data Analysis

- Developed an AI-driven automatic platform that unifies state-of-the-art R/Python-based single-cell workflows and demonstrated its utility through a cross-species lung cancer metastasis atlas and revealed a conserved metastatic program and microenvironmental adaptations.

Department of Immunology, Peking University 06/2022 – 07/2026

Undergraduate Research Intern, part-time

Advisor: Dr. Qing Ge

Sparse Integrative Modeling of Drug-Response States in Multiple Myeloma (MM)

- Developed PRISM-MM, a sparse integrative dictionary learning framework that disentangles shared drug-response programs from study-, drug-, and cell source-specific variation.
- Curated a large-scale MM drug-perturbation transcriptomic compendium from 37 GEO studies, comprising 800 samples across 16 drugs and 32 MM cell lines, together with held-out bulk and scRNA-seq datasets for model validation.
- Benchmarked PRISM-MM against representative perturbation-prediction models.
- Identified conserved drug-response programs and validated them across held-out independent bulk and scRNA-seq datasets, revealing recurrent shifts from plasma-cell secretory programs toward inflammatory and stress-tolerant states.
- Identified candidate state-maintaining genes and an *RFXAP*-centered immune-regulatory axis associated with a conserved inflammatory adaptation state, providing potential mechanisms underlying drug resistance in MM.

Multimodal Clonal and Transcriptomic Profiling of MGUS-to-MM Progression

- Implemented an integrative computational workflow combining BCR clonotype information with scRNA-seq to resolve clonal dynamics and transcriptional remodeling during plasma-cell disease progression.

Unveiling Single-Cell Transcriptomic Landscape and Molecular Pathogenesis of Tricuspid Regurgitation (TR)

- Combined TR patient-derived scRNA-seq with histological evidence to identify disrupted cell communication and profibrotic signaling in TR, and further validated these molecular alterations using public valve-disease datasets.

Developing a 3D Spheroid Model to Mimic the Microenvironment of Extramedullary MM Metastasis

- Identified upregulation of the mechanosensitive regulator *PIEZO1* in extramedullary lesions using spatial transcriptomics data of MM, established tunable hydrogel-based 3D spheroid models, and functionally validated *PIEZO1-YAP/TAZ* signaling as a mediator of stress-induced bortezomib resistance.

CONFERENCE PRESENTATIONS

1. **Symposium on Innovation Experimental Design, School of Basic Medical Sciences, Peking University** 12/2024
   Wu M, Xu D, Ge Q. Transcriptome Atlas of Human Tricuspid Valves Reveals Dysregulated Anti- and Pro-Inflammatory Cellular Interaction Involved in Tricuspid Regurgitation. (Poster Presentation)
2. **Growing to Shape Future Medicine: The 7th Student Innovation Conference, Peking University** 12/2023
   Xiao J*, Liu J*, Wang H*, **Wu M***, Xie Z*. Explore the Correlation Between Myopia Degree and Intensity/Time of Physical Activity in College Students. (Poster Presentation)

ACTIVITIES

Peking University Healthy E-sports Association
Co-founder & Vice President 09/2023 – 09/2025

- Co-founded the association and helped grow it to 120 members
- Organized company visits, inter-university e-sports competitions and competition-watching events

Student Union, School of Basic Medical Sciences, Peking University
Vice President of the Development Department 09/2022 – 09/2023

- Organized major campus events, including New Year Gala, Christmas Singing Event, and Mid-Autumn Mooncake Workshop, engaging more than 200 participants in total

Volunteer Service in the Local Community
Student Volunteer 06/2022 – 07/2022

- Distributed food and essential medical supplies to local residents during the COVID-19 pandemic
- Cleaned and maintained public facilities to support community hygiene and sanitation

Volunteer Tutor for College Entrance Exam
Chengdu No. 7 High School 08/2021 – 12/2021

- Tutored high school students in comprehensive science (physics, chemistry, and biology)
- Developed personalized study plans and provided exam strategies

PEER REVIEW SERVICE

Co-reviewer with Dr. Monkol Lek

Assessed manuscripts for ***Cell*** (07/2026).

Assessed manuscripts for ***Nature Medicine*** (08/2026).

SKILLS

Genomics: bulk/scRNA-seq, ATAC-seq, 3D genomics, spatial transcriptomics, WGS/WES, multi-omics integration

Computational: tool development (*SPADE*, *SCAPE*), machine-learning framework and model fitting (*PRISM-MM*)

Experimental: PCR, western blotting, ELISA, cell culture, flow cytometry, 3D spheroid models

Language: English (TOEFL 105/120; reading 28, listening 26, speaking 24 and writing 27), Chinese (native)